# Electronic Strain Measurement

At-home lab

## Executive summary

Firstly, well done! It is assumed that at this stage you have a functioning, tested, and calibrated instrumentation amplifier and strain measurement device!

For your third at home lab, you will further investigate simple bending theory, and observe the direct measurement of surface strain.

This laboratory incorporates the electronics taught to-date, with an everyday engineering measurement device: a strain gauge. In the discussion, you should comment on the advantages and limitations of this measurement technology from both observation and independent research.

The beam theory covered in section 8 of the Structures (Engineering Science) lecture notes is also worth comparing to the observation made from the experiments.

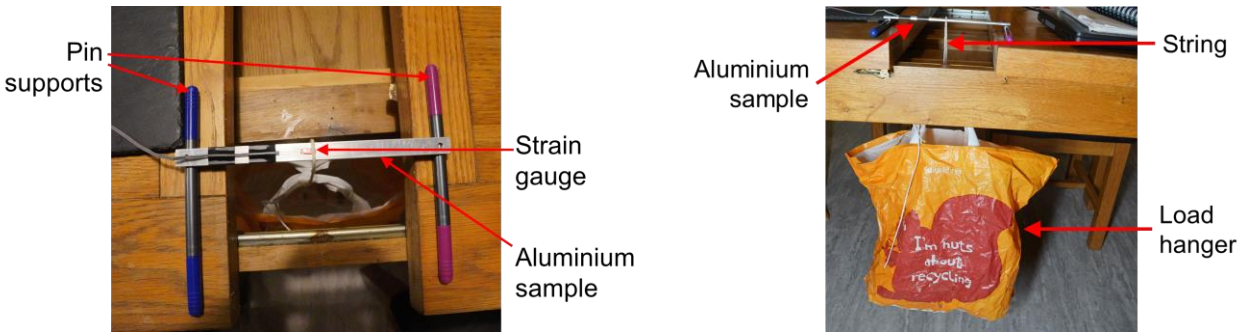
Please note that whilst this instructional note utilises the lab template in places the tone of English is not written as a lab report.

# 1. Introduction

If you are reading this lab instruction then you must have already:
1) Assembled your strain gauge (in your pair)
2) Calibrated your strain gauge, using the shunt method (in your pair)
This instruction now asks you to work individually to complete two experiments to test the aluminium specimen.

# 2. The bending experimental procedure

The first proposed test is in simple bending. The first proposed test is in simple bending. As before, safe use of ingenuity and creativity will form an important part of your experiments. A larger load will be required to develop measurable strain as the specimen has a relatively large cross-sectional area. It is suggested that up to 4 standard tin cans can be used. These will have a mass of approximately 2 kg. Care should be taken to ensure small children do not interact with the equipment and the load should be hung as close to the floor as possible to minimise the fall distance. Two photographs of a potential set up are shown in Figure 1.




(a) (b)

Figure 1 - Showing example setup of (a) pin supports and specimen and (b) elevation of experimental setup

Once an appropriate support system has been created, run through the following steps:

1. Position the supports 10 mm in from each end of the specimen.
2. Measure and record the span.
3. Position load hanger at mid span.
4. With only the mass of your load hanger applied, balance the strain gauge bridge using the potentiometer and pot screwdriver.
5. Weigh the first tin can (if you do not have a method for this and you are using 400g cans, assume 465g per can).
6. Apply 1 tin can of load to the specimen and record the voltage change.
7. Repeat steps 5 and 6, adding 1 tin can each time, until you have applied 4 tin cans.
8. Repeat the experiment a further 2 times, such that a mean can be found to reduce the influence of random errors.
9. Repeat the experiment with the strain gauge on the other side (flip the specimen over)
10. Finally, measure the breadth and depth of the specimen section. The Vernier callipers would be a good tool for this.

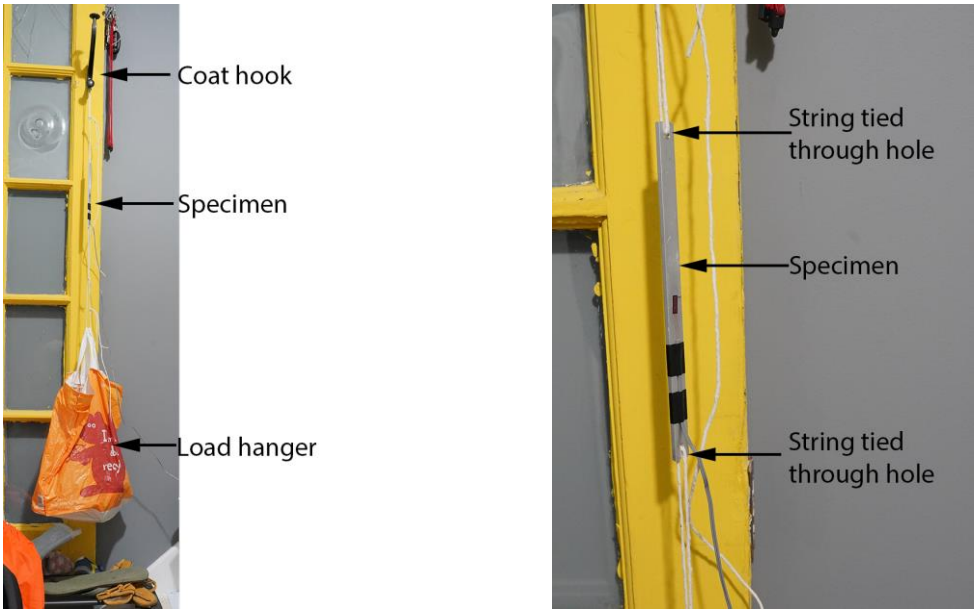
N.B. assume the elastic modulus of aluminium to be 70 GPa

## 3. The tension experimental procedure

Using the same safety advice as above, set up a tension test from a suitable datum.

A coat hook might be a good starting point.

As before, ensure the load is kept as close to the floor as possible. The proposed set up can be seen in Figure 2.




(a) (b)

Figure 2 - (a) Overview of tension set up, (b) close-up of specimen

*Hint – this experiment is intended to stretch the instrumentation and electronics to its measurement limit (not failure limit!). Please record the results you observe / measure with the multimeter and discuss in your discussion section.*

Once an appropriate support system has been created, run through the following steps:

1. With only the mass of your load hanger applied, balance the strain gauge bridge using the potentiometer and pot screwdriver.
2. Weight the first tin can (if you do not have a method for this and you are using 400g cans, assume 465g per can)
3. Apply 1 tin can of load to the specimen and record the voltage change
4. Repeat steps 2 and 3, adding 1 tin can each time, until you have applied 4 tin cans.
5. Once complete, repeat the experiment a further 2 times such that a mean can be found to reduce the influence of random errors.

## 4. Further observations

When taking measurements for both experiments, estimate the signal to noise ratio. This can be estimated by observing the sensor drift / noise when no load is applied and comparing this this to the maximum value achieved in the experiment.

Compare the measured strain with that of the theoretical values which can be obtained using beam

theory and stress definitions from Section 8.3 and 5.3 in structures (Engineering Science).

## 5. Report expectation

Ensure all the guidance received in session 1 regarding the template, format and report writing style, is implemented when writing up **Introduction, Methods, Results, Discussion and Conclusion** sections. No further sections of a report are required at this stage as we build up to a full report throughout the year.